Formal Framing: Geometric Routing via Riemannian–Complex Structure

1 Core Hypothesis

We propose that standard transformer architectures are constrained by their reliance on Euclidean linear operators, which inadequately capture directional, phase-dependent, and hierarchical structure.

We introduce a geometric routing framework in which:

- Local dynamics are governed by a complex-valued transport law
- Global structure is encoded in a Riemannian manifold
- Information propagation follows geodesic-aware routing rather than purely linear mixing

2 Geometric State Space

Let (M, g) be a smooth Riemannian manifold with metric g. Each token or latent state evolves as:

$$x_t \in M$$

3 Tangent Space Representation

At each point x_t, we define a local representation in the tangent space:

$$v_t \in T_{x_t}M$$

To introduce directional and phase-aware structure, we extend this to a complexified tangent space:

$$z_t \in T_{x_t}M \otimes \mathbb{C}$$

with decomposition:

$$z_t = v_t^{(r)} + i\, v_t^{(i)}$$

4 Local Transport Operator (Complex Phase Dynamics)

We define a local update operator:

$$\mathcal{U}_{x_t} : T_{x_t}M \otimes \mathbb{C} \;\to\; T_{x_t}M \otimes \mathbb{C}$$

which encodes:

- rotation (phase evolution)
- scaling (amplitude modulation)
- directional coupling

A minimal form:

$$\mathcal{U}_{x_t}(z) = W_r z + i\, W_i z$$

where W_r, W_i are learned linear maps on $T_{x_t}M$.

This generalizes real-valued linear layers by allowing intrinsic rotational structure rather than encoding it implicitly.

5 Manifold Update (Geometric Routing Step)

The updated state is obtained via exponential mapping:

$$x_{t+1} = \exp_{x_t}\Big(\Re\big(\mathcal{U}_{x_t}(z_t)\big)\Big)$$

Key properties:

- respects manifold geometry
- enforces constraint structure (e.g., curvature, boundedness)
- replaces unconstrained linear accumulation

6 Interpretation

This framework separates computation into two regimes:

Local (Tangent Space)

- complex-valued
- phase-aware
- linearizable
- supports rotation/interference

Global (Manifold)

- nonlinear
- curvature-aware
- enforces structure
- governs routing and compression

7 Relationship to Standard Architectures

Component	Standard Transformer	Proposed Framework
State space	$\mathbb{R}^n$	Riemannian manifold M
Local ops	real linear layers	complex transport operators
Mixing	attention / linear projection	geodesic-aware routing
Geometry	implicit / flat	explicit / curved

8 Key Claim

The advantage does not arise from complex numbers alone. Rather, it emerges from:

> The interaction between complex local transport (phase dynamics) and global Riemannian structure (geodesic routing), which induces non-linear compression and structured information flow not achievable in purely Euclidean linear systems.

9 Minimal Theoretical Statement

Let M be a Riemannian manifold and $T_xM \otimes \mathbb{C}$ its complexified tangent bundle. A routing process defined by local complex-linear transport followed by exponential mapping induces a nonlinear flow on M whose trajectories encode both phase-dependent dynamics and curvature-constrained propagation.

10 Intuition

- **Euclidean models:** information spreads like diffusion in a flat grid.
- **Proposed model:** information moves like waves constrained to a curved surface, with phase, interference, preferred directions, and natural compression into geometric sectors.

11 What "$2i$" Actually Becomes

- $i \rightarrow$ local 90° rotation operator
- $2i \rightarrow$ rotation + scaling

In this framework, scalar complex multiplication generalizes to a learned complex transport field over a manifold:

$$\text{scalar } 2i \quad \longrightarrow \quad \mathcal{U}_{x_t}(z) = W_r z + i\, W_i z$$